AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 8 · WORKBOOK

Structured Output and Schemas

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected structured-output audit.

Distinguish layers, audit fields/types, inspect source support and keep permissions separate.

Keep one habit: Retain original text, exact schema, source identity, first failure and evidence limits.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Prompt as a Contract. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from A Distribution of Possibilities. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Iterative Prompting

Prompt as a Contract

A Distribution of Possibilities

W1 · Distinguish the layers

> State what parsing, contract checks and source comparison establish.

Your evidence record

- Room schema: exactly status,room,capacity,source_ids. status supported or needs_review; source_ids distinct known strings. supported requires nonempty room text and nonnegative exact integer capacity, excluding Boolean. needs_review requires null room and capacity. Match approved current room and capacity sources after schema.

  Separate book picture uses title,pages,source_id and record R River Notes64; do not mix fields.

- Explain what parsing, **schema** checks and source support each establish.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Audit types and fields

> Null, missing, Boolean and string values are different.

Your evidence record

- Room schema: exactly status,room,capacity,source_ids. status supported or needs_review; source_ids distinct known strings. supported requires nonempty room text and nonnegative exact integer capacity, excluding Boolean. needs_review requires null room and capacity. Match approved current room and capacity sources after schema.

  Separate book picture uses title,pages,source_id and record R River Notes64; do not mix fields.

- Why do string "12",boolean true,missing capacity and an extra action **field** violate this contract?

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Check what a cited source supports

> An existing ID does not establish the associated value.

Your evidence record

- Room schema: exactly status,room,capacity,source_ids. status supported or needs_review; source_ids distinct known strings. supported requires nonempty room text and nonnegative exact integer capacity, excluding Boolean. needs_review requires null room and capacity. Match approved current room and capacity sources after schema.

  Separate book picture uses title,pages,source_id and record R River Notes64; do not mix fields.

- Sources P,Q support B capacity 12. A **schema**-valid object cites P,Q but gives 99. Which layer rejects it?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Keep report validity separate from permissions

> A valid read-only report cannot authorise booking.

Your evidence record

- Room schema: exactly status,room,capacity,source_ids. status supported or needs_review; source_ids distinct known strings. supported requires nonempty room text and nonnegative exact integer capacity, excluding Boolean. needs_review requires null room and capacity. Match approved current room and capacity sources after schema.

  Separate book picture uses title,pages,source_id and record R River Notes64; do not mix fields.

- A fully valid supported reporting object is returned. May the application automatically book the room? Explain.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Validate new objects independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Treat the independent X/Y/Z descriptions as properly encoded JSON objects with exactly status,room,capacity,source_ids; the shorthand is a description, not literal JSON. Source A is the current approval for F and B the current register for F capacity9. Apply exact integer capacity and nonempty room rules before matching source support. Do not change G to F or text9 to integer9 silently.

Sources A approve room F and B register F capacity 9. Object X has supported,F,9,[A,B] with exactly the required **field**s. Y replaces 9 with "9". Z keeps integer 9 but names room G.

Trace the first failing layer for each and explain the difference.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Validate new objects independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Require answer, source snippet, citation, uncertainty, and next-question **field**s.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Audit **type**s and **field**s for Structured **output**s and **validation**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Structured **output**s and **validation**. Proposed explanation: “Valid JSON means correct answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Schema requires title:text and pages:number. **Output** is {"title":"Cloud","pages":"80"}. Fix its structural error. Would the repaired object prove the book really has 80 pages?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Schema requires title text and pages number. **Output** is {"title":"River","pages":"80"}.

Diagnose the shape error.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A required **schema** has title as text and count as an integer. The **output** is {"title":"Seeds","count":"three"}. Identify the error and give a valid replacement.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 8 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Schema requires title:text and pages:number. **Output** is {"title":"Cloud","pages":"80"}. Fix its structural error. Would the repaired object prove the book really has 80 pages?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Sources A approve room F and B register F capacity 9. Object X has supported,F,9,[A,B] with exactly the required **field**s. Y replaces 9 with "9". Z keeps integer 9 but names room G.

Trace the first failing layer for each and explain the difference.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

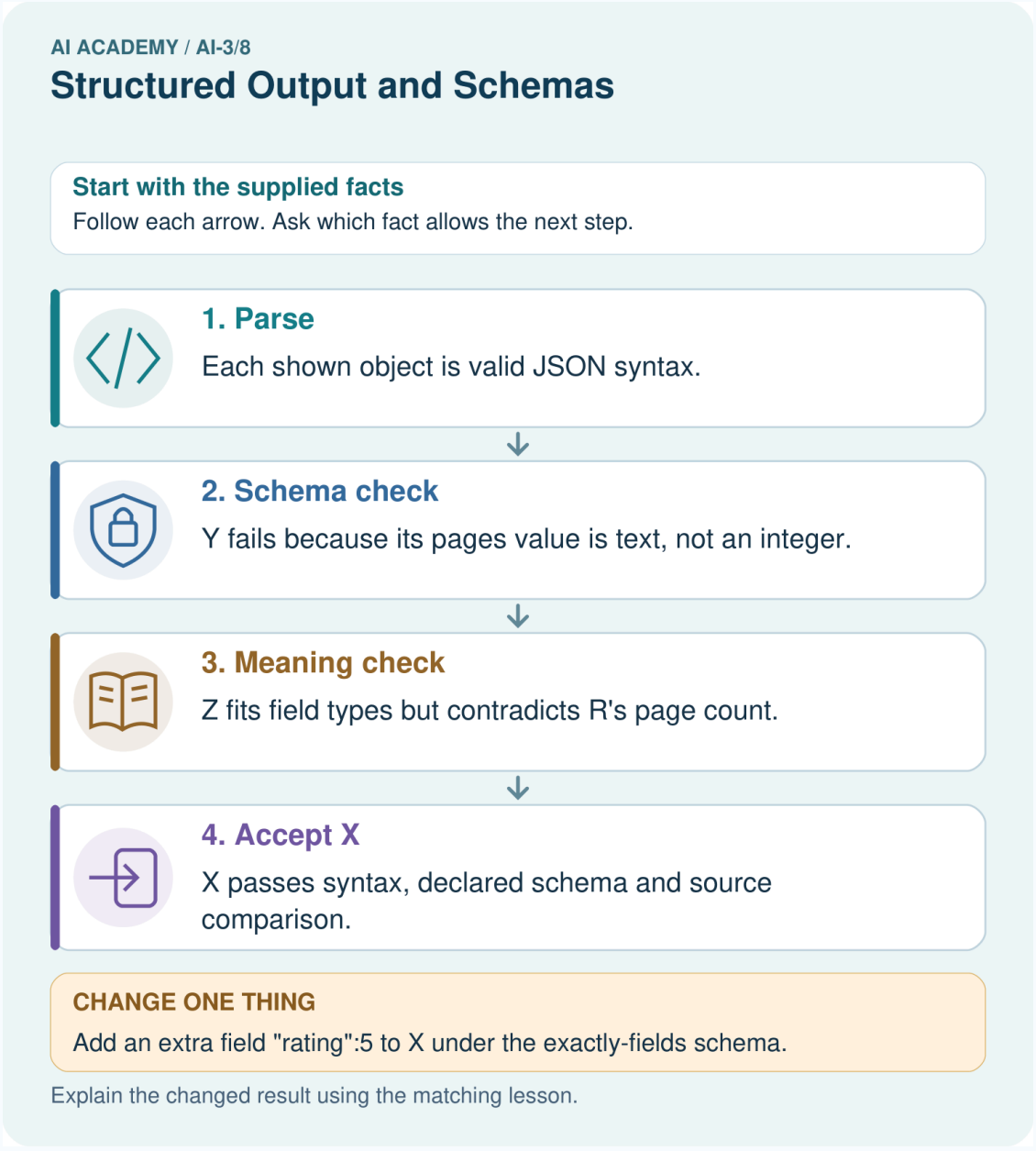
> Keep the explanation and the support you used.

Use this when helpful

A fictional Study Buddy returns a machine-readable book fact.

Schema: exactly title text, pages integer, source_id text. Approved record R: River Notes, 64 pages.

X={"title":"River Notes","pages":64,"source_id":"R"}; Y uses pages:"64"; Z uses pages:46 with otherwise identical **field**s.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which object demonstrates that valid JSON and valid **type**s are insufficient for truth?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Validate Study Buddy **output**s before another program consumes them; retain source IDs and separate syntax/**type**/support failures.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Validate Study Buddy **output**s before another program consumes them; retain source IDs and separate syntax/**type**/support failures.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional Study Buddy returns a machine-readable book fact.
- Which labelled step matches this detail? X passes syntax, declared **schema** and source comparison.
- 1. Meaning check
- 2. Accept X

- 3. **Schema** check
- 4. Parse
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Accept X X passes syntax, declared **schema** and source comparison. Parsing establishes that text forms a structured object.

Schema checks establish required **field**s and **type**s. **Semantic validation** compares their meaning with the approved source. These are different layers: a parseable, well-shaped object can still contain a false fact.

Report the first failing layer without treating format success as factual success.

Changed-case question

- Changed-case practice: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The changed object fails **schema** despite keeping the correct page count.
- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The changed object fails **schema** despite keeping the correct page count. Exactly means additional **field**s are not allowed in this contract.

Repair and save your work

Review Iterative Prompting (ai-3/7). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: What **evidence** supports improvement here, and what remains unproven?.

Return to this week and explain what you changed.

Review Prompt as a Contract (ai-3/5). Goal: Write a prompt specification with observable response criteria. Short practice: Why should the tool not fill melting from memory even if it knows a common definition?.

Return to this week and explain what you changed.

Review A Distribution of Possibilities (ai-3/2). Goal: Check whether a next-token table is a valid probability distribution. Short practice: Why is 0.625 not a broken probability sum?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

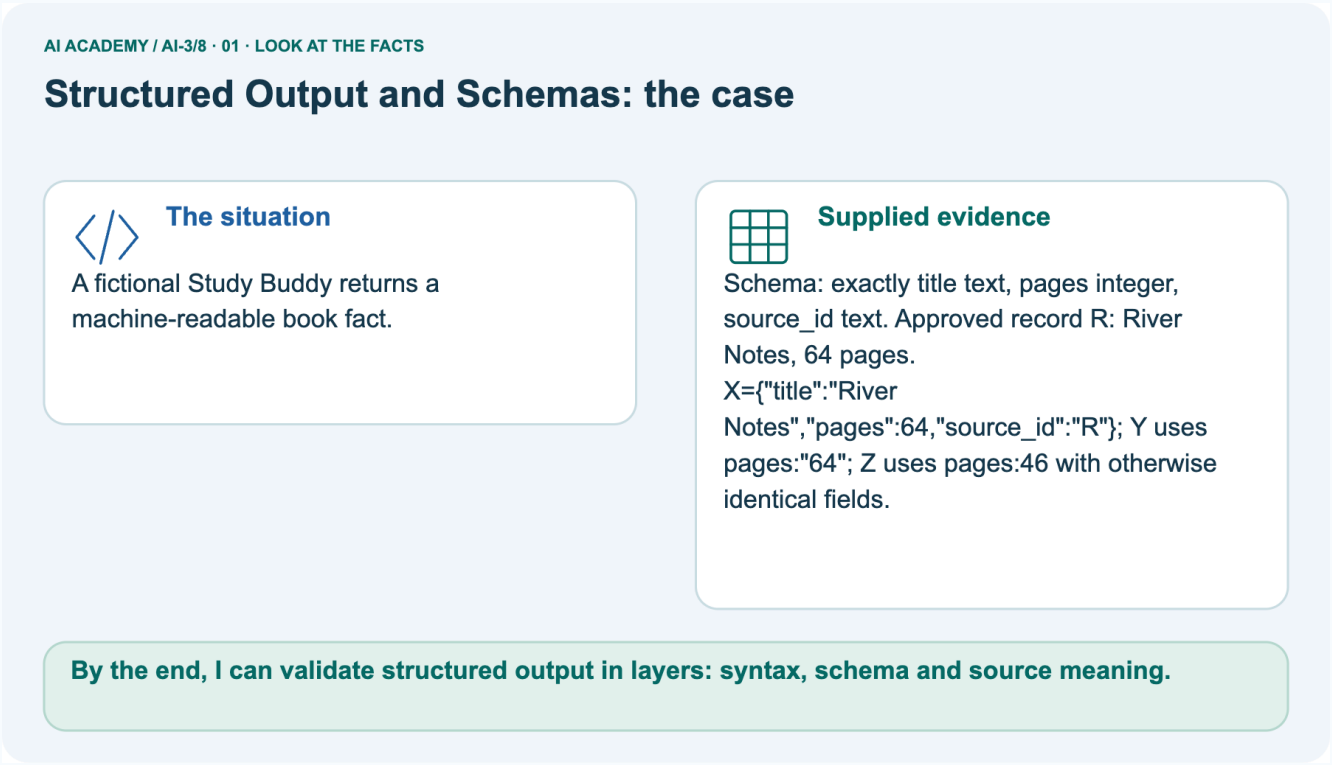
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Structured Output and Schemas: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

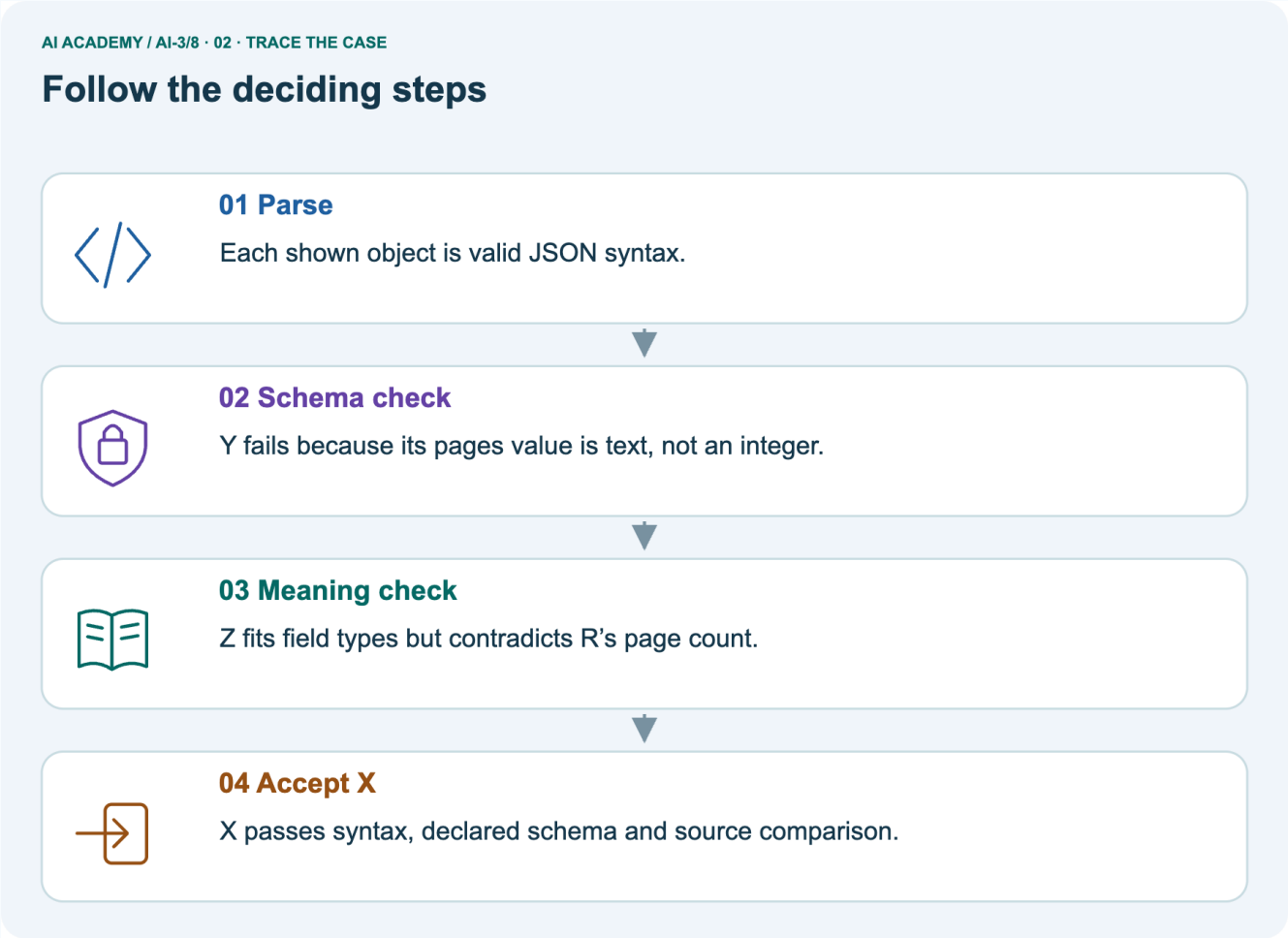
A fictional Study Buddy returns a machine-readable book fact. Supplied facts: **Schema**: exactly title text, pages integer, source_id text. Approved record R: River Notes, 64 pages.

X={"title":"River Notes","pages":64,"source_id":"R"}; Y uses pages:"64"; Z uses pages:46 with otherwise identical **field**s.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



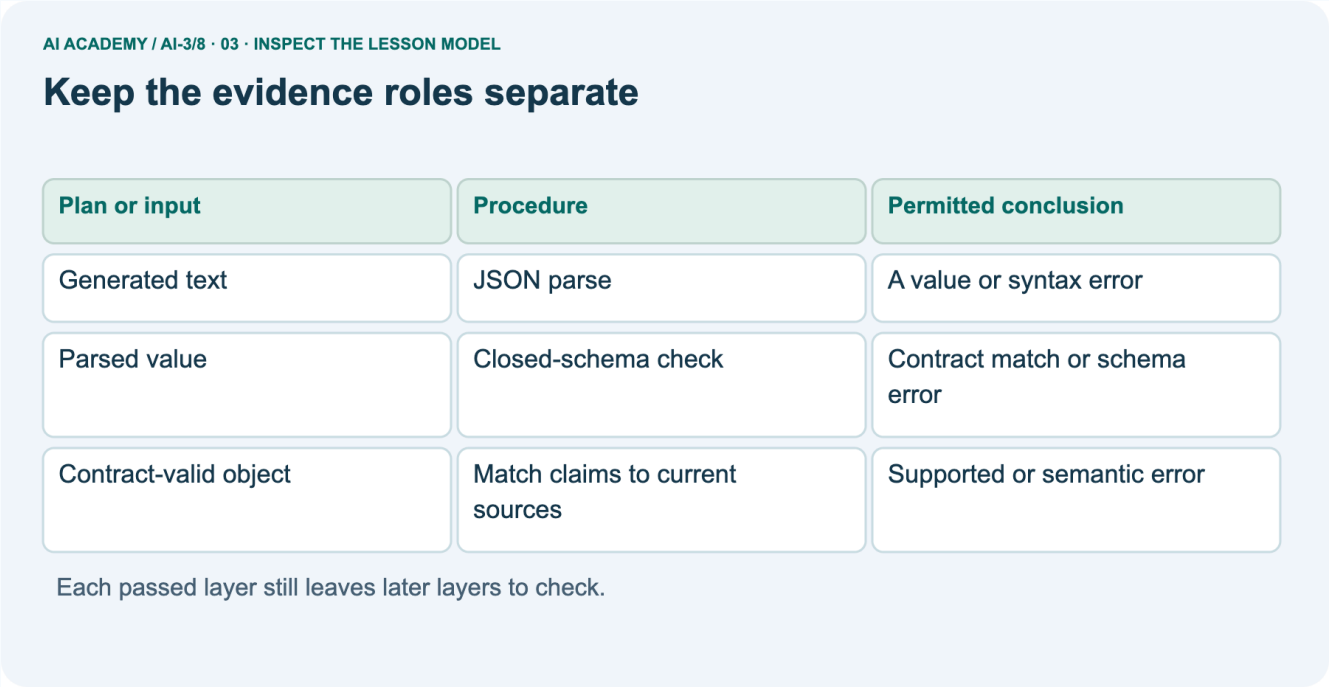
Follow the labelled picture with the matching explanation. Open the vector diagram

Parse: Each shown object is valid JSON syntax. → **Schema** check: Y fails because its pages value is text, not an integer. → Meaning check: Z fits **field type**s but contradicts R's page count. → Accept X: X passes syntax, declared **schema** and source comparison.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep the evidence roles separate



Follow the labelled picture with the matching explanation. Open the vector diagram

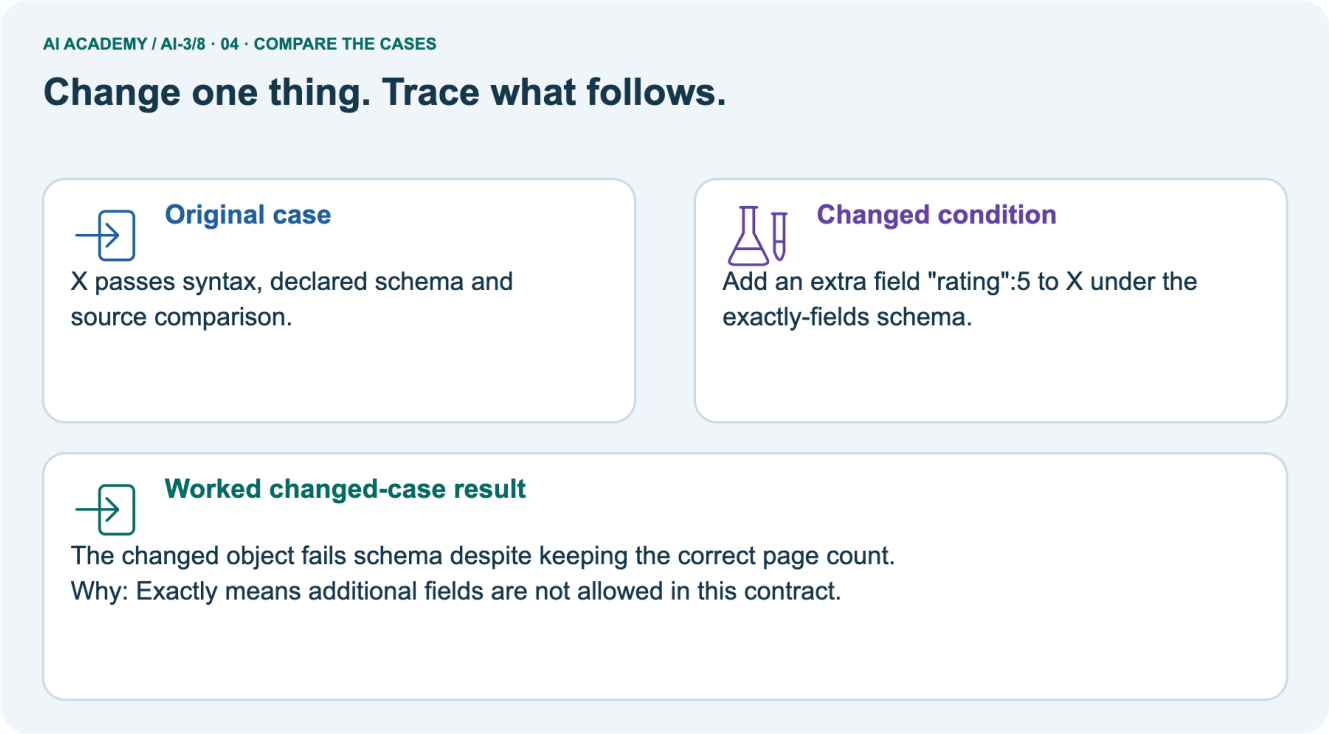
Keep the **evidence** roles separate. Generated text | JSON parse | A value or syntax error. Parsed value | Closed-**schema** check | Contract match or **schema** error.

Contract-valid object | Match claims to current sources | Supported or semantic error

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

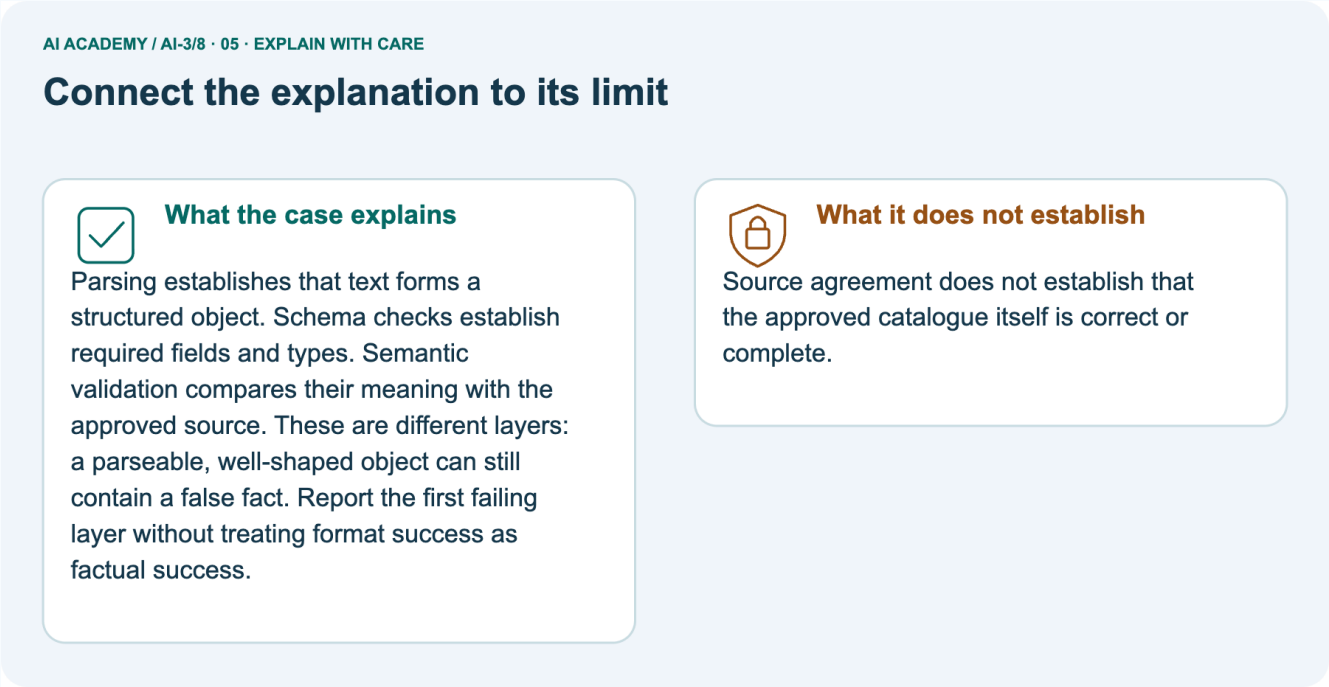
Original case: X passes syntax, declared **schema** and source comparison. Changed condition: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**.

Expected result: The changed object fails **schema** despite keeping the correct page count. Reason: Exactly means additional **field**s are not allowed in this contract.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Parsing establishes that text forms a structured object. **Schema** checks establish required **field**s and **type**s. **Semantic validation** compares their meaning with the approved source.

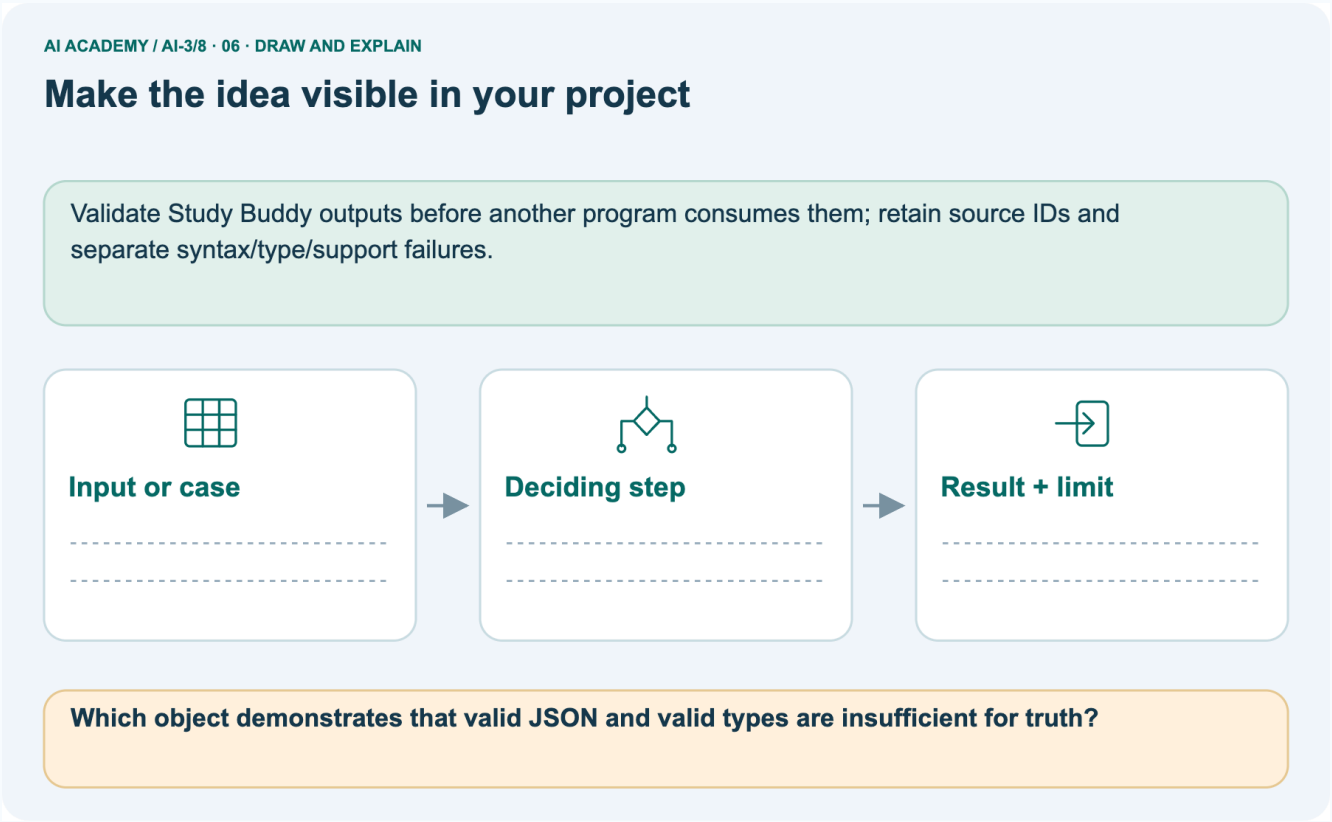
These are different layers: a parseable, well-shaped object can still contain a false fact. Report the first failing layer without treating format success as factual success.

Limit: Source agreement does not establish that the approved catalogue itself is correct or complete.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate Study Buddy **output**s before another program consumes them; retain source IDs and separate syntax/**type**/support failures. Draw your **input**, deciding step, result and limit.

Supported question: Which object demonstrates that valid JSON and valid **type**s are insufficient for truth?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Structured Output and Schemas: the case

02. Follow the deciding steps

03. Keep the evidence roles separate

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.